

## **Identified issues and initial website Review**

### **Overview**

The HTML structure provided was relatively basic and contained a limited number of validation errors. In accordance with the project requirements, all HTML pages (index.html, about.html, projects.html, and contact.html) were validated using the W3C Markup Validation Service.

The validation process identified several issues, most notably the absence of the lang attribute within the root <html> element and missing alt attributes on image elements. These findings aligned with observations already noted during the initial review.

The lang attribute is important for accessibility and assists browsers and screen readers in correctly interpreting the page content. Similarly, the alt attribute provides descriptive text for images, improving accessibility and ensuring users are still informed when images fail to load.

Furthermore, a div element with the class name header was present; however, no semantic <header> element had been included within the page structure. This has been corrected to improve semantic HTML and accessibility.

An “Images” folder was also created and incorporated into the project structure, as image assets had not originally been included.

## **HTML Improvements Implemented**

### **1. index.html**

The following corrections and enhancements were implemented:

- Added the lang attribute to the <html> element.
- Added a semantic <nav> element.
- Added navigation links to the remaining pages.
- Added a semantic <header> element.
- Added a semantic <main> element.
- Added a semantic <footer> element.
- Relocated the footer outside of the main content area.
- Added descriptive alt attributes to all images.

- Added required meta tags within the <head> section.

## **2. about.html**

The following corrections and enhancements were implemented:

- Added the lang attribute to the <html> element.
- Added a semantic <nav> element.
- Added navigation links to the remaining pages.
- Added a semantic <header> element.
- Added a semantic <main> element.
- Added a semantic <footer> element.
- Relocated the footer outside of the main content area.
- Added descriptive alt attributes to all images.
- Added required meta tags within the <head> section.

## **3. projects.html**

The following corrections and enhancements were implemented:

- Applied the same structural improvements as those made to index.html and about.html.
- Added a semantic <section> element to group project content.
- Replaced generic <div> elements with semantic <article> elements where appropriate.
- Added unique id attributes to all article elements to improve navigation and identification.

## **4. contact.html**

The following corrections and enhancements were implemented:

- Applied the same structural improvements as those made to the other pages.
- Added labels to all form input fields.
- Added radio button input fields.

- Added checkbox input fields.
- Added required validation to the Name and Email fields.
- Added minimum and maximum length validation to the Telephone field.
- Improved overall form accessibility and usability.

## **CSS Review and Issues Identified**

The stylesheet contained multiple structural, accessibility, and styling issues. The following issues were identified:

### **1. Invalid HTML Tags within the CSS File**

HTML tags (`<br><br>`) were included directly within the CSS file. HTML markup should not be present in the CSS.

### **2. Additional Invalid HTML Tag**

The tag `<br aria-hidden="true">` was also included in the CSS file and should be removed.

### **3. Missing Navigation Styling**

No styling rules were provided for the navigation menu.

### **4. Header Alignment**

The `.header` class lacks appropriate alignment styling, resulting in inconsistent presentation.

### **5. Poor Colour Contrast**

The `.hero` section uses a light blue background which may negatively impact readability and accessibility for some users.

### **6. Accessibility Considerations**

Image accessibility was not fully addressed. While this is primarily handled within the HTML through alt attributes, it was noted as a missing consideration.

### **7. Limited Use of CSS Selectors**

The stylesheet uses only a small number of selector types and does not demonstrate a sufficient range of selector techniques.

### **8. Missing Pseudo-Class Selectors**

Pseudo-classes such as the following were not implemented and they were completely left out:

- `:hover`
- `:focus`
- `:active`

## **9. Missing Combination Selectors**

More advanced selectors were not demonstrated, for example:

`.header > h1`

,

`form input[type="text"]`

## **10. Incomplete Form Styling**

Form inputs currently only utilize display and margin properties. Additional styling should be applied to improve usability and appearance.

## **11. Missing Box Model Demonstration**

The stylesheet does not clearly demonstrate the use of following, and has been added :

- Margin
- Padding
- Border

as part of the CSS box model.

## **12. Missing Display Property Demonstration**

There is no clear example showcasing the differences between:

- Block elements
- Inline elements
- Inline-block elements

## **13. Limited Text Styling**

The stylesheet does not make use of additional typography properties such as:

- `font-weight`
- `text-transform`
- `text-decoration`

- letter-spacing

#### **14. Limited Colour Styling**

Colour usage is minimal and lacks variety throughout the design.

#### **15. Table Styling Not Implemented**

No styling exists for table elements despite table styling being a required CSS competency.

#### **16. Incorrect Footer Alignment**

The footer currently uses:

text-align: left;

A centred alignment would provide a more balanced and professional presentation.

#### **Additional Resources Included**

To improve the overall presentation and functionality of the website, an “Images” folder was added to the project containing:

- Website logo
- Hero image
- Project showcase images

These assets were integrated into the relevant pages.